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Neighborhood deprivation and mortality among people with Alzheimer's disease and related dementia

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Abstract

Background: There are limited studies quantifying the extent to which neighborhood deprivation affects mortality in individuals living with Alzheimer's disease (AD) and AD-related dementias (AD/ADRD).

Objective: To quantify to what extent neighborhood deprivation affects adverse outcomes in individuals living with AD/ADRD.

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Ethical considerations

Ethical approval was obtained by the internal review board at the University of Florida, Gainesville, FL, USA.

Declaration of conflicting interests

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Methods: We identified individuals with AD/ADRD using a 15% random sample of national Medicare fee-for-service beneficiaries. Area deprivation index (ADI) was spatially linked to the Medicare AD/ADRD cohort using zip codes. Multivariate logistic regression was applied to examine the association between ADI and all-cause mortality among beneficiaries with AD/ADRD.

Results: After adjusting for patients characteristics, compared to the lowest ADI quartile (Q1), higher ADI quartiles were associated with higher odds of all-cause mortality in individuals with AD/ADRD (Q2 = OR 1.08 [95% CI: 1.06, 1.10], Q3=OR 1.09 [95% CI: 1.07, 1.11], and Q4 OR 1.05 [95% CI: 1.03, 1.07]).

Conclusions: Neighborhood deprivation is an independent risk factor for mortality in persons with AD/ADRD.

Keywords

Alzheimer's disease; area deprivation index; mortality; social determinants of health

Introduction

Alzheimer's disease and Alzheimer's disease-related dementias (AD/ADRD), as coded by the International Classification of Diseases version 10 (ICD-10), are a class of fatal progressive neurodegenerative diseases that present cognitive impairment and limitations in daily living. Currently, over six million people are living with AD/ADRD in the United States with a projection of 12.7 million by 2050.[1] Individuals ≥65 years of age survive an average of four to eight years after a diagnosis of AD/ADRD.[1] Meeting the complex care needs of people living with dementia is challenging. Furthermore, people living with dementia from racial-ethnic minoritized groups, with poorer access to insurance, and in socioeconomically disadvantaged environments are more likely to encounter barriers to diagnosis and care.[2]

Clinical and social risk factors are linked to the causes and outcomes of AD/ADRD.[3] Understanding the risk factors associated with adverse outcomes among AD/ADRD patients and the development of interventions to address these outcomes is of high public health relevance due to the direct influence of health outcomes on quality of life and health care utilization and expenditures. Among these risk factors, a significant portion are non-medical, highlighting the need to examine the broader context of health. This brings us to the concept of social determinants of health (SDoH).

The World Health Organization defines SDoH as "*non-medical factors that influence health outcomes. They are the conditions in which people are born, grow, work, live and age…*".[4] This suggests that SDoH encompass fundamental contextual factors that correlate with exposure to adversity at a population level.[5] A fundamental component of upstream SDoH is neighborhood-level deprivation.[6]

Research has found that neighborhood-level factors can predict the onset of chronic conditions, mortality, and physical function.[7] Neighborhood deprivation, therefore, may be a suitable target for interventions and policy programs to improve structural equity and population health.[6,8,9] A growing body of work has evaluated the associations between neighborhood SDoH and cognitive health and AD/ADRD onset.[10,11] Besser et al. and Wu et al. reported an association between economically disadvantaged neighborhoods and cognitive decline in their independent systematic reviews.[12,13] However, limited research has assessed the extent of neighborhood deprivation impact on health outcomes among individuals living with AD/ADRD in the United States.

The objective of this study was to evaluate the associations between neighborhood deprivation and the risk of mortality among older adults living with AD/ADRD. We hypothesized that higher levels of neighborhood deprivation were associated with higher odds of all-cause mortality in individuals living with AD/ADRD even when accounting for person-level demographic and clinical characteristics. We also conducted subgroup analyses to identify the potential interaction effect of neighborhood deprivation and individual sociodemographic factors in association with mortality in patients living with AD/ADRD.

Methods

Study design and population

The current study used data from a 15% random sample of the National Medicare Beneficiary database for the available insurance claims data between January 1, 2016, and December 31, 2018. The study cohort included Medicare fee-for-service beneficiaries diagnosed with AD/ADRD on or before January 1, 2017 (index date). AD/ADRD diagnosis was identified using the AD/ADRD indicator from the Center for Medicare and Medicaid Services (CMS) Chronic Condition Warehouse (CCW).[14] Beneficiaries were excluded for any of the following reasons: died before the index date; had non-continuous enrollment during the baseline year (January 1, 2016, to December 31, 2016); did not have Medicare fee-for-service enrollment information during the baseline period; any enrollment information (managed care); or had missing data on area information.

To ensure complete medical data, only those with continuous fee-for-service enrollment were included. Managed care enrollees were excluded due to limited access to detailed individual-level healthcare utilization and diagnosis data.

The follow-up period for this study began on January 1, 2017, and ended on either the participants' date of death or December 31, 2018.

Outcome

The outcome variable was all-cause mortality, identified during the follow-up between January 1, 2017, and December 31, 2018 through CMS data.

Exposure

The area deprivation index (ADI) is a measure of neighborhood-level deprivation. ADI is a validated neighborhood-level composite index of 17 SDoH measures derived from the US Census surveys and American Community Survey (ACS) data.[9] The four domains that encompass these measures include: education, income/employment, housing, and household characteristics.[15] ADI is scored between 1 to 10 with the least disadvantaged neighborhood conditions designated by lower scores and the greatest deprivation indicated by higher scores.[9] For the current study, ADI, which was measured in 2015, was operationalized into quartiles. The lowest quartile indicated the lowest deprivation group. Additionally, ADI and the Medicare AD/ADRD cohort were spatially linked using 5-digit zip codes in 2016–2018.

Covariates

Sociodemographic and clinical characteristic covariates were extracted on or before the index date from the Medicare claims data for analyses. Sociodemographic characteristics included the following: age, sex (male or female), racial/ethnic group (Non-Hispanic White, Non-Hispanic Black, Hispanic, Asian/Pacific Islander, American Indian/Alaskan Native, and Other), disability status (identified based on Medicare enrollment file), Medicare low-income subsidy recipient (individuals with income of up to 150% of the Federal Poverty Level), Medicaid enrollment, and rural/urban residence.

Participants were also classified as living in either urban or rural areas using the 9-level United States Department of Agriculture (USDA) Rural-Urban Continuum Codes (RUCC).[16] Location was dichotomized to urban-dwelling participants (levels 1–3) and rural participants (levels 4–9), which is consistent with the USDA classifications. Clinical characteristics included duration since the AD/ADRD diagnosis (identified using CMS Chronic Condition Warehouse indicators that trace the first diagnosis of the conditions date back to January 1, 1999) and having a history of the following diseases or conditions: acute myocardial infarction, atrial fibrillation, stroke, diabetes, chronic kidney disease, depression, and CMS priority types of cancer. Indicator variables were defined for each condition using CMS Chronic Condition Warehouse indicators (the indicator equaled 1 if the first diagnosis happened prior to the index date, 0 otherwise).

Statistical analysis

Participant characteristics were summarized using means (standard deviations) or counts (percentages). The primary analysis quantified the association between ADI and allcause mortality, using multivariable logistic regression models. Models were adjusted for age, sex, racial/ethnic group, urbanicity (urban/ rural), low-income subsidy enrollment, baseline AD/ADRD year, duration since AD/ADRD diagnoses, acute myocardial infarction, chronic kidney disease, depression, diabetes, and trans-ischemic stroke, cancer, and chronic heart failure. ANOVA and Post hoc Tukeys' HSD tests assessed differences between ADI quartiles for race/ethnicity, urbanicity, low-income subsidy, Medicaid status, mortality outcome, duration of AD/ADRD, acute myocardial infarction, cancer, congestive heart failure, chronic kidney disease, depression, diabetes, and transient ischemic stroke. Potential non-independence between participants within the same zip codes were evaluated using generalized estimating equations (GEE).

Secondary subgroup analyses were conducted by urbanicity, Medicaid enrollment (yes/no), and racial/ethnic groups to identify whether the association between neighborhood deprivation and mortality varied by individual-level sociodemographic factors. Medicaid coverage and race/ethnicity were included in this secondary analysis, given that neighborhood deprivation-mortality may vary by these factors. Adjusted odds ratios (aOR) and 95% confidence intervals of the resulting models were reported. Interactions between ADI and each of the subgroups (urbanicity, race, and Medicaid status) were evaluated, and rates of death were also considered in the secondary analyses by ADI quartile and each subgroup (urbanicity, race, and Medicaid status), respectively. All statistical analyses were performed using SAS 9.4 statistical software (SAS Institute, Cary, NC) and used an alpha level of 0.05 to evaluate statistical significance.

Results

A cohort of 651,466 participants was included in the final analyses (see participant flowchart in Figure 1). In the analytic dataset, the mean age was 79.4 (±11.6), 65% were female, 74.8% white, and the mean AD/ADRD duration was 4.6 ±4.1 years (Table 1). Additionally, 28.3% of patients died over the median follow-up period of 360 days (range: 30–729 days).

The ANOVA analysis for patient characteristics and medical history shows that ADI quartiles differ significantly regarding race/ethnicity, urbanicity, low-income subsidy, Medicaid status, mortality outcome, duration of AD/ADRD, and background medical comorbidities. In post hoc Tukeys' HSD tests, we found that ADI quartiles all statistically differ at a 0.05 level, save between the following comparisons: Diabetes and Congestive heart failure (ADI Q(1) –ADI Q(2), > 0.05), Depression (ADI Q(3)-ADIQ(4), $p > 0.05$), and AD/ADRD mortality (ADI Q(1)-ADI Q(4), $p > 0.05$) (Table 1).

A GEE model was used to evaluate the association between AD/ADRD mortality and ADI quartile to account for clustering by ZIP code. Compared to the first ADI quartile, all other ADI quartiles had increased odds of ADRD/AD mortality (ADI Q(2) OR: 1.09, $p < 0.0001$. ADI Q(3) OR: 1.09 $p < 0.0001$, ADI Q(4) 1.04, $p = 0.001$) (Table 2).

In this cohort, there were significantly higher odds of all-cause mortality for participants in all three higher ADI quartiles when compared to the first quartile (Q1), with an aOR 1.08 [95% CI: 1.06, 1.10] for 2nd quartile (Q2), aOR 1.09 [95% CI: 1.07, 1.11] for the third quartile (Q3), and 1.05 [95% CI: 1.03, 1.07] for the fourth quartile (Q4) (Table 3), after adjusting for individual-level sociodemographic and clinical characteristics.

Interaction terms between ADI quartiles and urbanicity, ADI quartiles and race/ethnic groups, and ADI quartiles and Medicaid enrollment were $p = 0.01$, $p < 0.0001$, and $p < 0.0001$, respectively. Subgroup analysis for beneficiaries identified as living in urban areas found that there were significantly higher odds of all-cause mortality in all three higher ADI quartiles when compared to the Q1 group, with an aOR 1.08 [95% CI: 1.06, 1.10] for Q2, aOR 1.09 [95% CI: 1.07, 1.11] for Q3, and aOR 1.06 [95% CI: 1.04, 1.09] for Q4. Conversely, in the rural subgroup, ADI did not show a significant association with mortality (compared to the Q1 of ADI, aORs 1.06 [95% CI: 0.98, 1.15] for Q2, 1.03 [95% CI: 0.96, 1.11] for Q3, and 0.96 [95% CI: 0.90, 1.03] for Q4 (Table 4).

Analysis by racial/ethnic groups indicated that there were significantly higher odds of all-cause mortality in all the higher ADI quartiles for the White and Asian groups. The Black racial group, however, had significantly higher odds of all-cause mortality in Q3 aOR 1.09 [95% CI: 1.03, 1.16], and marginally in Q2 and Q4 (aORs 1.05 [95% CI: 0.98, 1,12], and

1.03 [95% CI: 0.97, 1.09], respectively). There was no significant ADI-mortality association within the American Indian/ Alaskan Native group (p value > 0.05) (Table 5). Conversely, the Q2, Q3, and Q4 of ADI were associated with reduced odds of all-cause mortality in the Hispanic group compared to the Q1 (aORs for Q2, Q3, and Q4 were 0.94 [95% CI: 0.89, 0.99], 0.92 [95% CI: 0.87, 0.98], and 0.91[95% CI: 0.86, 0.96], respectively).

Subgroup analysis, stratified by Medicaid enrollment status, showed a similar association between ADI and allcause mortality in both groups, but the association was stronger among those with Medicaid enrollment (Table 6).

Rates of death were higher for all ADI quartiles amongst Medicaid recipients compared to non-recipients. In comparison of urban and rural status, mortality rate was highest within the 2nd and 3rd ADI quartiles amongst the rural population, and the rural population exhibited higher overall rates of death compared to the urban population. Between race/ethnicity groups, compared to the non-Hispanic White, all Race/Ethnicity groups exhibited lower rates of death in ADI quartiles, save American Indian/Alaskan Natives in the first ADI quartile. Rate of death did not appear to differ within the Black race group across ADI quartile, while rate of death decreased for Asian/Pacific Islander population as ADI level increased. Oppositely, Hispanic populations experienced higher rates of death at higher ADI quartiles.

Discussion

Neighborhood SDoH refers to the broader social and built environmental factors, within a community or region, that influence health outcomes. SDoH significantly influences both environmental factors affecting health and the biological processes within the body. These processes, driven by ongoing interactions between our surroundings and our biology, can lead to long-term negative health outcomes by causing changes in areas like our genes, brain development, and immune system, ultimately increasing the risk of morbidity and mortality.[17,18] These factors are increasingly recognized as a vital targets for developing healthcare policies designed to improve population health management and value-based care. In the current study, we used nationally representative data to examine the impact of neighborhood SDoH on the health outcomes of patients with AD/ADRD. In support of our hypothesis, greater neighborhood deprivation was associated with greater odds of all-cause mortality. We observed a 5–9% increase in the odds for all-cause mortality in the Q2-Q4 compared to the Q1 ADI quartile, respectively. Moreover, we identified that ADI is a significant risk factor for mortality in patients with AD/ADRD, specifically in urban settings.

There is a current dearth of research examining the associations between social, demographic, and deprivation risk factors at a neighborhood- level, and health outcomes in persons with AD/ADRD.[19,20] Prior research has found that economic deprivation, defined by economic assets or income, was associated with increased mortality rates in individuals with AD/ADRD.[17,18] For example, a Netherlands-based study demonstrated that lower SES was associated with higher absolute 1 and 5 year mortality risk in patients with dementia.[21] Other studies, however, found no associations between income and mortality risk in people with dementia. Overall, many of these existing studies are limited to small sample sizes and homogenous study populations.[21–24] Previous research that utilized a multifactorial index for neighborhood deprivation (e.g., 2019 Indices of Multiple Deprivation and UK Index of Multiple Deprivation) determined that higher levels of area deprivation were associated with increased incidence of mortality amongst those living with dementia.[25–27]

The current study evaluated the impact of neighborhood-level risk factors on all-cause mortality and found evidence suggesting that ADI is a risk factor for adverse health outcomes in individuals with AD/ADRD living in urban but not rural areas. These findings may be explained by the differences in the population distribution of ADI in urban and rural areas. The ADI values were approximately normally distributed in rural residents, with a large variation ranging from 5% to 58%. Of note, the rural reference group (Q1) was relatively smaller than the other quartiles. In contrast, the ADI value distribution was more homogenous, with less variation (ranging from 19% to 29%) in urban areas.

Previous research evaluated the association of area deprivation in rural areas on health outcomes for chronic diseases and found differences between ADI regions regarding high quality diabetes care and medication adherence.[28] A study assessing financial toxicity and the capacity for sustained treatment in breast cancer had suggested that financial burden may be independent of rurality, while still being influenced by area deprivation.[29] Combined influence of high area deprivation, geographic rurality, and socioeconomic strain may create significant hurdles in accessing and achieving sustained care for these conditions for patients who require long term care management in individuals with progressive diseases, such as AD/ADRD.

Other studies have noted that post-hospitalization, depression rates differ by ADI quartile, with areas of higher deprivation experiencing greater rates of depression.[30] These findings, when considered alongside our study's results on ADI and mortality, may speak to a broader burden of managing progressive diseases within more deprived regions.

Medication adherence and air pollution may help explain the differences between and within urban and rural ADI groups.[31,32] Pilonieta et al., found that living in higher deprived areas were associated with a lower probability of medication adherence in patients with AD/ADRD.[33] This medication non-adherence and low adherence increases the risk of mortality in older adults.[34]

Regarding air pollution, Smargiassi et al. found that levels of exposure to PM2.5 and NO2 —both linked to increased incidence AD/ADRD— were associated with distances to major roads, a factor that often coincides with living in more deprived areas.[35,36] In addition, increased exposure to air-pollutants ($PM_{2.5}$, PM_{10}, NO_2, and black carbon) is associated with all-cause mortality.[37] Further research is needed to investigate these suggested explanations within the AD/ADRD population.

We found a similar trend that higher ADI quartiles were associated with increased odds of mortality when compared to the groups living in the least deprived neighborhoods for White, Black, and Asian groups. In part, non-significant results could be attributed to insufficient statistical power, as observed in the American Indian/ Alaskan Native and Other groups.

A possible reason, though poorly understood, for the significant ADI-mortality in the White but not Black group is the “Black-White mortality crossover”. This idea posits that mortality is higher in White persons at older ages when compared to Black persons at older ages.[38] There are two prevailing explanations: The first theory postulates that an inaccurate reporting of age on death certificates of which occurs more commonly in disadvantaged groups. Estimates are typically biased downwards at the oldest ages in these disadvantaged groups.[38,39] The second theory posits that “differential heterogeneity in frailty” leads to the mortality crossover; disadvantaged groups experience higher mortality rates at younger ages and therefore leaving a more robust older surviving population.[38,40] Therefore, the total population composition is weighted towards a more robust disadvantaged group that have lower mortality rates when compared to white racial groups.[38,40]

In addition, disparities in diagnosis and access to healthcare of these groups and those in higher ADI locations, may help explain our findings. Evidence has shown that diagnosis of AD/ADRD occurs at a later stage of the disease for these disadvantaged groups. Possible rationale being higher racial and ethnic disparities in underdiagnoses and misdiagnoses, in addition to accessing diagnostic services later in their illness.[41–43]

Interestingly, we observed an inverse association between ADI and mortality in Hispanic individuals with AD/ADRD. The “Hispanic mortality paradox” may partly explain this observation; prior research has found that there is lower mortality in Hispanic/Latino groups despite low socioeconomic status and educational attainment. This mechanism may help explain the protective effect and significant associations of decreased mortality found in this study, where 29.3% of the Hispanic/Latino group lives in the highest quartile. Cultural differences and greater familial care may provide another explanation for the null associations between deprivation groups within the minority groups.[25,44]

This study makes a significant contribution by evaluating the associations of neighborhood deprivation and all-cause mortality using a cohort of Medicare beneficiaries with AD/ADRD with spatially linked ADI data. However, there are several limitations. We could not obtain important clinical data such as dementia severity, duration of cognitive decline, and individual-level SDoH, which can be important confounding factors in the association between neighborhood deprivation and mortality in AD/ADRD patients. Of note, though we have duration of diagnosis, cognitive problems may have developed before diagnosis. Furthermore, this study relied on a place of residence at the time of diagnosis and therefore could not accurately assess SDoH exposure throughout the participant’s lifetime. The strengths of this study include the fact that it has a nationally representative dataset of individuals with AD/ADRD. Taken together, this study further contextualizes the implied structural inequity found at a neighborhood level for individuals with AD/ADRD.

Conclusion

This study quantified the role of neighborhood deprivation in mortality risk after a diagnosis of AD/ADRD and found that ADI is a significant risk factor for mortality in individuals with AD/ADRD even after adjusting for individuals’ demographic and clinical characteristics. Our findings suggested structural inequities are associated with health outcomes among millions of Americans with dementia. Future research should prioritize developing and accessing health policy and intervention programs to address neighborhood deprivation-associated disparities and ultimately improve health equity.

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Data availability statement

The data that support the findings of this study are available from Centers for Medicare & Medicaid Services but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. Data are however available from the Centers for Medicare & Medicaid Services upon request and permission.

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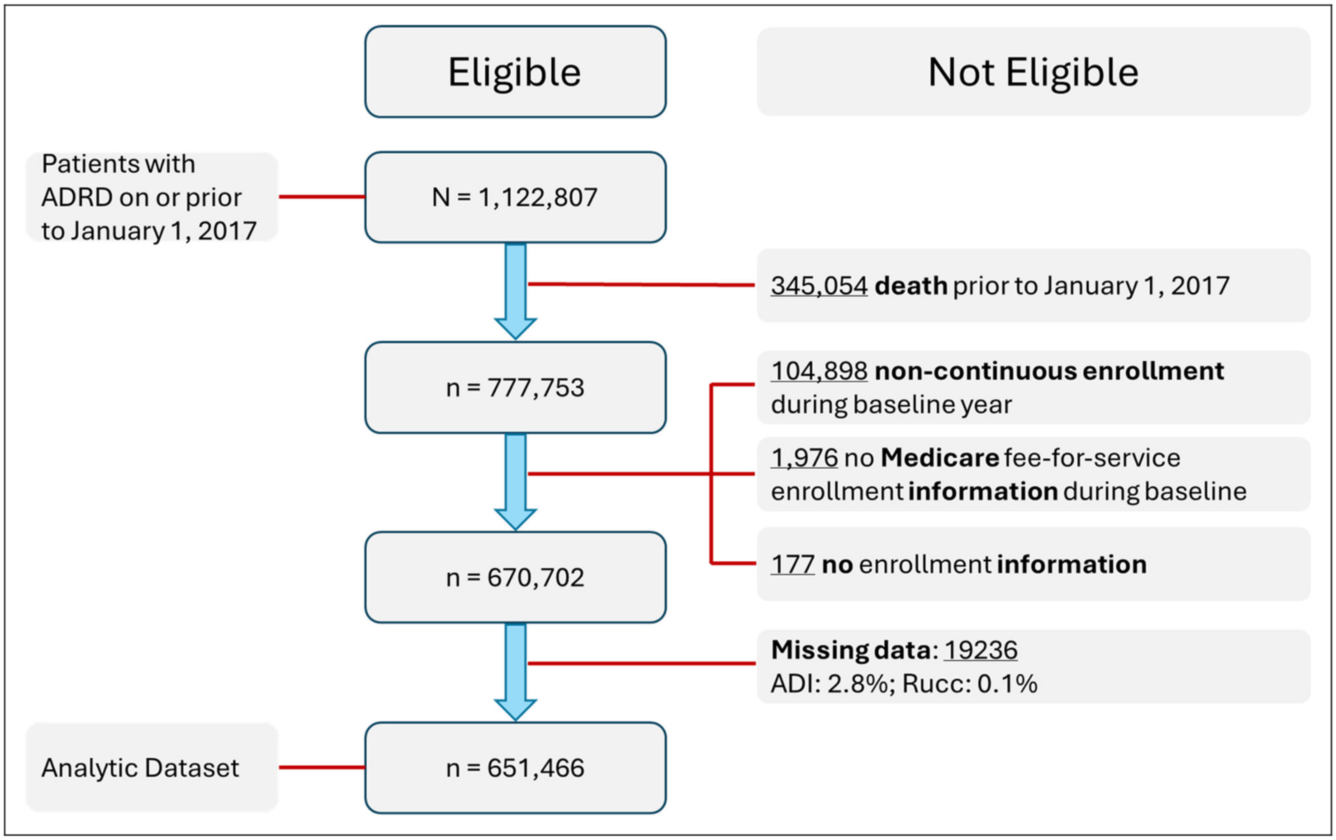

Figure 1.
Participant flow chart.

Table 1.

Patient characteristics by area deprivation index quartile.

Characteristics	Area deprivation index quartiles*				Comparison by area deprivation quartiles**	
	1st, (n=162,897)	2nd, (n=162,801)	3rd, (n=162,866)	4th, (n=162,902)	F-statistic (F) (df)	P***
Age, mean (sd)	81.09 (10.83)	79.87 (11.32)	78.95 (11.80)	78.00 (12.02)	2144.91 (3)	<0.0001
Female, % (n)	64.89 (105709)	65.45 (106552)	65.70 (106997)	65.07 (105995)	9.51 (3)	<0.0001 ***
Race, % (n)						
White	79.13 (128897)	77.23 (125726)	75.77 (123398)	67.20 (109470)	309.54 (3)	<0.0001 ***
Black	5.83 (9498)	7.96 (12965)	10.53 (17147)	18.08 (29455)		
Hispanic	7.92 (12901)	11.52 (18760)	11.42 (18604)	12.80 (20854)		
Asian/Pacific Islander	5.50 (8953)	2.10 (3419)	1.23 (2003)	0.78 (1274)		
American Indian/Alaska Native	0.12 (193)	0.22 (358)	0.31 (508)	0.55 (900)		
Other	1.51 (2455)	0.97 (1573)	0.74 (1206)	0.59 (949)		
Urban, % (n)	97.25 (158417)	93.62 (152418)	83.89 (136628)	65.64 (106923)	21966.5 (3)	<0.0001 ***
Low Income Subsidy, % (n)	38.19 (62206)	44.02 (71662)	51.00 (83060)	59.49 (96906)	5644.69 (3)	<0.0001 ***
Medicaid, % (n)	36.71 (59798)	41.92 (68243)	48.28 (78631)	55.92 (91098)	4609.26 (3)	<0.0001 ***
Mortality Outcome, % (n)	27.84 (45357)	28.64 (46623)	28.82 (46940)	28.11 (45792)	16.53 (3)	<0.0001 ***
Medical History, % (n)						
Duration of AD/ADRD (y), mean (sd)	4.76 (4.15)	4.62 (4.06)	4.65 (4.09)	4.68 (4.09)	32.48 (3)	<0.0001
Acute Myocardial Infarction	7.34 (11962)	7.82 (12732)	8.32 (13557)	9.01 (14684)	111.61 (3)	<0.0001 ***
Atrial Fibrillation	26.50 (43171)	25.22 (41054)	24.49 (39880)	22.97 (37421)	189.81 (3)	<0.0001 ***
Cancer	20.49 (33384)	18.73 (30490)	17.78 (28955)	17.03 (27738)	242.09 (3)	<0.0001 ***
Congestive Heart Failure	47.56 (77476)	47.63 (77534)	49.24 (80190)	52.39 (85345)	333.87 (3)	<0.0001 ***
Chronic Kidney Disease	48.67 (79288)	51.58 (83967)	52.81 (86010)	55.01 (89700)	464.01 (3)	<0.0001 ***
Depression	65.21 (106221)	67.99 (110685)	69.56 (113289)	69.75 (113631)	331.66 (3)	<0.0001 ***
Diabetes	51.98 (84668)	52.37 (85251)	53.98 (87908)	57.81 (94171)	465.92 (3)	<0.0001 ***
Transient Ischemic Stroke	35.39 (57647)	36.01 (58617)	36.97 (60207)	37.86 (61678)	82.93 (3)	<0.0001 ***

*Area Deprivation quartiles have the following rangers: 1st(0–25%), 2nd (26–50%), 3rd (51–75%), 4th (76–100%).

** Comparison of the variables by area deprivation quartiles was assessed using ANOVA and post-hoc Tukey's Test.

*** Tukey's test significant at the 0.05 level.

Table 2.

Generalized estimation equation assessing ADI quartile and covariates toward odds of AD/ADRD mortality, accounting for ZIP code as repeated subject.

Parameter	Odds ratio	95% CI	P
ADI_Q (1)	ref	ref	ref
ADI_Q (2)	1.09	(1.07, 1.12)	<0.0001
ADI_Q (3)	1.09	(1.06, 1.11)	<0.0001
ADI_Q (4)	1.04	(1.01, 1.06)	0.001
Race (White)	ref	ref	ref
RACE (Black)	0.79	(0.77, 0.81)	<0.0001
Race (Asian/Pacific Islander)	0.55	(0.54, 0.57)	<0.0001
Race (Hispanic)	0.52	(0.50, 0.55)	<0.0001
Race (American Indian/Alaska Native)	0.90	(0.80, 1.00)	0.05
Race (other)	0.67	(0.69, 0.72)	<0.0001
Urban	0.94	(0.92, 0.96)	<0.0001
ADRD_AF_2010	1.04	(1.01, 1.07)	0.002
Duration AD/ADRD	0.99	(0.99, 0.99)	<0.0001
Female	0.72	(0.71, 0.73)	<0.0001
Age	1.07	(1.06, 1.07)	<0.0001
Medicaid	1.61	(1.57, 1.65)	<0.0001
LIS	1.03	(0.99, 1.07)	0.10
AMI	1.22	(1.20, 1.24)	<0.0001
CKD	1.37	(1.36, 1.39)	<0.0001
Depression	1.23	(1.21, 1.24)	<0.0001
Diabetes	0.99	(0.97, 1.00)	0.02
Stroke	1.15	(1.13, 1.16)	<0.0001
Cancer	1.13	(1.12, 1.15)	<0.0001
CHF	1.45	(1.43, 1.47)	<0.0001

Table 3.

Association of neighborhood deprivation and death by quartiles of area deprivation index in the full analytic sample (N=651,466).

Quartile	N	Odds ratio (95% CI)
1st	162897	reference
2nd	162801	**1.08 (1.06, 1.10)**
3rd	162866	**1.09 (1.07, 1.11)**
4th	162902	**1.05 (1.03, 1.07)**

*Adjusted for age, sex, race and ethnicity, rural/urban residence, low-income subsidy status, Medicaid enrollment, year of AD/ADRD diagnosis, and history of the following: atrial fibrillation, acute myocardial infarction, cancer, chronic heart failure, chronic kidney disease, depression, diabetes, and transient ischemic stroke.

Bolded figures indicate a statistical significance of $p \leq 0.05$.

Table 4.

Association of neighborhood deprivation and death by quartiles of area deprivation index in urban and rural areas.

	Quartile	N	Odds ratio (95% CI)
Urban (N=554,386)	1st	158417	reference
	2nd	152418	**1.08 (1.06, 1.10)**
	3rd	136628	**1.09 (1.07, 1.11)**
	4th	106923	**1.06 (1.04, 1.09)**
Rural (N=97,080)	1st	4480	reference
	2nd	10383	1.06 (0.98, 1.15)
	3rd	26238	1.03 (0.96, 1.11)
	4th	55979	0.96 (0.90, 1.03)

* Adjusted for age, sex, race and ethnicity, low-income subsidy status, Medicaid enrollment, year of AD/ADRD diagnosis, and history of the following: atrial fibrillation, acute myocardial infarction, cancer, chronic heart failure, chronic kidney disease, depression, diabetes, and transient ischemic stroke.

Bolded figures indicate a statistical significance of $p \leq 0.05$.

Table 5.

Association of neighborhood deprivation and death by quartiles of area deprivation index by racial/ethnic group.

	Quartile	N	Odds ratio (95% CI)
White (N=487,491)	1st	128897	reference
	2nd	125726	**1.10 (1.08, 1.12)**
	3rd	123398	**1.10 (1.08, 1.12)**
	4th	109476	**1.07 (1.05, 1.09)**
Black (N=69,065)	1st	9498	reference
	2nd	12965	1.05 (0.98, 1.12)
	3rd	17147	**1.09 (1.03, 1.16)**
	4th	29455	1.03 (0.97, 1.09)
Asian/Pacific Islander (N=15,649)	1st	8953	reference
	2nd	3419	**1.15 (1.04, 1.28)**
	3rd	2003	**1.25 (1.11, 1.42)**
	4th	1274	**1.27 (1.09, 1.47)**
Hispanic (N=71,119)	1st	12901	reference
	2nd	18760	**0.94 (0.89, 0.99)**
	3rd	18604	**0.92 (0.87, 0.98)**
	4th	20854	**0.91 (0.86, 0.96)**
American Indian/Alaska Native (N=1959)	1st	193	reference
	2nd	358	0.86 (0.57, 1.31)
	3rd	508	0.92 (0.62, 1.37)
	4th	900	0.92 (0.62, 1.35)
Other (N = 6183)	1st	2455	reference
	2nd	1573	1.12 (0.95, 1.32)
	3rd	1206	**1.49 (1.25, 1.78)**
	4th	949	1.18 (0.96, 1.44)

*Adjusted for age, sex, rural/urban residence, low-income subsidy status, Medicaid enrollment, year of AD/ADRD diagnosis, and history of the following: atrial fibrillation, acute myocardial infarction, cancer, chronic heart failure, chronic kidney disease, depression, diabetes, and transient ischemic stroke.

Bolded figures indicate a statistical significance of $p \leq 0.05$.

Table 6.

Association of neighborhood deprivation and death by quartiles of area deprivation by medicaid recipient status.

	Quartile	N	Odds ratio (95% CI)
Receives Medicaid (N=297,770)	1st	59798	reference
	2nd	68243	**1.12 (1.09, 1.15)**
	3rd	78631	**1.16 (1.14, 1.19)**
	4th	91098	**1.05 (1.02, 1.08)**
Does Not Receive Medicaid (N=353,696)	1st	103099	reference
	2nd	94558	**1.06 (1.03, 1.08)**
	3rd	84235	**1.04 (1.01, 1.06)**
	4th	71804	1.02 (0.99, 1.04)

* Adjusted for age, sex, race and ethnicity, rural/urban residence, low-income subsidy status, year of AD/ADRD diagnosis, and history of the following: atrial fibrillation, acute myocardial infarction, cancer, chronic heart failure, chronic kidney disease, depression, diabetes, and transient ischemic stroke.

Bolded figures indicate a statistical significance of $p \leq 0.05$.